**Regis University**
**College for Professional Studies**
**School of Computer & Information Sciences**

# SYLLABUS

## COURSE NUMBER: CS 450

## COURSE TITLE: Data Networks

### Course Description

This course introduces the concepts and terminology of data communications and network design. Course topics include transmission techniques, network topologies, protocols, security, network control, and network architectures.

### Course Prerequisites

CS 208 Computer Science Fundamentals
CS 362 Control Structures

### Course Outcomes

Upon successful completion of this course, students should be able to:

1. Provide a unified overview of the broad field of data and computer communications,
2. Identify the basic principles of data communications and network design that unify this field, such as multiplexing, flow control, and error control,
3. Recognize alternative approaches to meeting specific communication requirements,
4. Explain how standards play an increasingly dominant role in this field,
5. Identify the requirements for data communications such as encoding, signaling, synchronization, media, and protocols,
6. Describe the need for security in the data communications environment and the solutions used to achieve secure communications, and
7. Explain the business need for interoperability and describe how internetworking devices and protocols are used to accomplish that end.
8. Design, implement, test, and validate code elements used in support of data networks.

## Course Materials and Resources, including Technology Tools

**Required**:

*Textbook*

Kurose, J. F., & Ross, K. W. (2013). Computer networking: A top-down approach (6th ed.). Boston, MA: Addison Wesley (Pearson). ISBN13: 978-0-13-285620-1 or ISBN10: 0-13-285620-4 hardcover.

**Required Hardware/Software:**

1) Access to a PC-compatible computer system running Windows 2000, NT or XP.
2) IP Subnet Calculator; can download from:
http://www.wildpackets.com/products/free_utilities/ipsubnetcalc/overview
3) A C++ Compiler: Download Dev-C++ (free)
4) A Java Compiler: http://java.oracle.com
5) Wireshark Packet Analyzer: www.wireshark.org
6) Wireshark Users Guide. http://www.wireshark.org/docs/wsug_html

The compilers will be used to complete the programming assignments. Other compilers can be used, but the facilitator will be most familiar with the compilers listed above, and will most likely use the compiler to grade your programs. If a student chooses to use another compiler, it is the student's responsibility to obtain any help needed to use the software, and to insure that all assigned programs compile correctly on the compiler.

**Supplemental:**

Palmer, M. and Sinclair, R.B. (2003). *Guide to Designing and Implementing Local and Wide Area Networks*. (2nd ed.). Boston: Thompson Course Technology. ISBN 10: 061912122X or ISBN 13: 9780619121228.

Marcus, J. S. (1999). *Designing Wide Area Networks and Internetworks*. Upper Saddle River: Addison-Wesley. ISBN 10: 0201695847 or ISBN 13: 9780201695847.

Stallings, W., (2007). *Data and Computer Communications*. (9th edition). Upper Saddle River, NJ: Prentice-Hall; ISBN 10: 0131392050 or ISBN 13: 9780131392052.

*Week 1*

Java Applets that reinforce the subjects discussed in Chapter 1 of the textbook.

- http://media.pearsoncmg.com/aw/aw_kurose_network_2/applets/transmission/delay.html
- http://media.pearsoncmg.com/aw/aw_kurose_network_2/applets/queuing/queuing.html

- http://media.pearsoncmg.com/aw/aw_kurose_network_2/applets/message/messagesegmentation.html

Wireshark video tutorial.

http://media-2.cacetech.com/video/wireshark/introduction-to-wireshark/

*Week 3*

Java Applets that reinforce the subjects discussed in Chapter 3 of the textbook.

- http://media.pearsoncmg.com/aw/aw_kurose_network_4/applets/go-back-n/index.html
- http://media.pearsoncmg.com/aw/aw_kurose_network_4/applets/SR/index.html
- http://media.pearsoncmg.com/aw/aw_kurose_network_4/applets/flow/FlowControl.htm
- http://media.pearsoncmg.com/aw/aw_kurose_network_4/applets/fairness/index.html

*Week 5*

Java Applets that reinforce the subjects discussed in Chapter 5 of the textbook.

http://media.pearsoncmg.com/aw/aw_kurose_network_2/applets/csmacd/csmacd.html

## Method of Instruction

This course presents material using a classroom lectures or online readings, textbook readings, and facilitated group discussions. Students are assessed via individual assignments. The facilitator may assign a final exam.

## Related Standards

| Related Standards | |
|---|---|
| | This program enables students to achieve by the time of graduation: |
| | An ability to apply knowledge of computing and mathematics appropriate to the discipline |
| | An ability to analyze a problem, and identify and define the computing requirements appropriate to its solution |
| | An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs |
| | An understanding of professional, ethical, legal, security and social issues and responsibilities |
| | An ability to communicate effectively with a range of audiences |
| | An ability to analyze the local and global impact of computing on individuals, organizations, and society |

| | |
|---|---|
| | An ability to use current techniques, skills, and tools necessary for computing practice |
| | An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices |
| | An ability to apply design and development principles in the construction of software systems of varying complexity |
| **Regis Nine Educational Outcomes** | |
| | Content Knowledge in Discipline |
| | Diverse Cultures, Perspectives, and Belief Systems |
| | Arts, Sciences, and Humanities |
| | Critical Thinking |
| | Communication |
| | Technology |
| | Ethical and Social Responsibility |
| | Leadership in Service to Others |
| | Lifelong Learning |
| **Jesuit Themes - These are themes that are embedded as part of our interactions and instruction at Regis, but are not written in as activities** | |
| | Magis, Finding God in All Things, Unity of Hearts and Mind, Contemplatives in Action, Care of the Whole Person, Men and Women for Others |

## First Week Assignment

- READ ***Chapter 1*** in the text.
- Read pages 80-81 in the Wireshark Users Guide. http://www.wireshark.org/docs/wsug_html
- Be PREPARED to ASK QUESTIONS on unclear areas in the assigned reading.

## Course Overview Grid

| Week | Learning Topics | Activities | Reading Assignments | Assignments |
|---|---|---|---|---|
| 1 | Data Communication & Networking Concepts | | Textbook, Ch. 1 (pp. 1-67) Wireshark (pp 80-81, 82-84 | Assignment #1 |
| 2 | Application Interfact (API) Concepts | | Textbook, Ch. 2 (83-168) (179-180) Wireshark lab (192, 193-195 | Assignment #2 |
| 3 | Transport Layer Concepts | | Textbook, Ch. 3 (pp. 185-284) Wireshark | Assignment #3 |

| | | | | |
|---|---|---|---|---|
| | | | (312, 313-314) | |
| 4 | Network Protocols Concepts | | Textbook, Ch 4. (pp. 305-412, 438-439) | Assignment #4 |
| 5 | Link Layer & Local Area Network Concepts | | Textbook, Ch 5 (pp. 433-501) Wireshark (520, 521-522) | Assignment #5 |
| 6 | The Physical Layer Concepts | | Textbook, Ch 6 (pp. 513-578) Etherreal Lab 59, 594-596 | Assignment #6 |
| 7 | Security and Internetworking Concepts | | Textbook, Ch. 8 (671-743, 753) Textbook, Ch. 9 (pp. 771-800) 803-804 | Assignment #7 |
| 8 | Multimedia Network Concepts | | Textbook, Ch. 7 (587-655) Assign. pp 682 684-686 | Assignment #8<br><br>(Final Exam)** |

First night assignment is due on the first day of class, other assignments at the end of the week (online) or the beginning of the next class.

** Your facilitator may assign a final exam.

## Student Evaluation Grid

| Assignment | Value (percent of overall course grade)** | |
|---|---|---|
| Participation | 10% | 10% |
| Assignments 1-6 | 11% | 10% |
| Assignments 7-8 | 12% | 10% |
| Final** | – | 10% |
| | | |
| **TOTAL** | **100 %** | |

**Your facilitator may assign a final exam. If a final is assigned, the percentages are taken from the right-hand column, otherwise the left-hand column is used.

## Regis University Grading Scale

| | | | |
|---|---|---|---|
| A | 93 – 100 | C | 73 - 76 |
| A- | 90 – 92 | C- | 70 – 72 |
| B+ | 87 – 89 | D+ | 67 – 69 |
| B | 83 – 86 | D | 63 – 66 |
| B- | 80 – 82 | D- | 60 – 62 |

| | | | |
|---|---|---|---|
| C+ | 77 – 79 | F | 59 or Lower |

**Attendance/Participation**
Facilitator will provide specific attendance/participation policies in accordance with the official attendance policy of the School of Computer and Information Sciences.

**Attendance Policy of the School of Computer and Information Sciences**
Student participation is essential. Students are responsible for all material covered in each week. Students' interaction is central to the learning experience and imparts critical observations and knowledge to all concerned.

If a student misses more than one week of class participation without prior faculty approval, the facilitator may assign a failing grade.

**Late Assignments/Assignment Revisions**
Facilitator will provide specific late assignments/assignment revisions policies not to exceed the maximum official policy of the School of Computer and Information Sciences.

**Late Assignments/Assignment Revisions Policy of the School of Computer and Information Sciences**
Occasionally, work, health, and family obligations have priority over course work. If you experience these issues, please notify the facilitator in advance. If an assignment is due, please make every effort to submit your assignment on time. In the case of a negotiated absence or delay, assignments turned in by the agreed upon due date will receive full credit.

If you have not negotiated with the facilitator, assignments turned in late will be graded, and then reduced by 3% per day. Assignments **will not** be accepted later than one week after due date. Any Week 8 assignments may only be 3 days late.

**Incomplete Grade**
An incomplete grade is given at the discretion of the facilitator. A facilitator is not required to assign an incomplete grade. Each occurrence will be negotiated on an individual basis.

**Incomplete Grade Policy of the College of Professional Studies**
A grade of Incomplete or 'I" denotes that the required work for the course is incomplete due to unforeseen circumstances. Unforeseen circumstances mean, for example, that an accident, an illness, a death, or a major life transition has occurred. This grade is awarded at the discretion of the instructor and is submitted with an alternate grade of "I/F."

A student must submit a written request to the instructor asking that an incomplete grade be assigned prior to the end of the term in which the course is taken. The course instructor will determine the length of time for course completion, and the remaining requirements to complete the course; however the maximum time for completion is the end of the following semester.

Faculty will notify students in writing of the approval/denial of their request and provide instructions and guidelines required to resolve the incomplete grade. If the course work is not

completed by the end of the next applicable semester and a grade change form submitted by the instructor, the incomplete grade reverts to the alternate grade assigned by the instructor and is calculated in the grade point average.

**Tutoring Information:**
Occasionally students need additional assistance with course content. Tutorial assistance is available to students through SmartThinking, writing assistance and personal tutoring. SmartThinking provides every student with ten hours of free online tutoring in writing, math, statistics, economics and accounting each year. Writing assistance is available in the form of Roving Writing Tutors and a variety of writing workshops. Individual personal tutors are available in a variety of discipline areas with fees and arrangements made between the individual student and tutor. For access to these services, go www.regis.edu, Current Student, Services for CPS Students, Academic Tools, Tutoring/SmartThinking.

**Disability Statement:**
Equal access to education means equal opportunity to learn in an academic setting. Under Section 504 of the 1973 Rehabilitation Act and the Americans with Disabilities Act of 1990, students with documented disabilities who are otherwise qualified have a right to equal access to classes at Regis University. If you have a documented disability requiring academic adjustments for this class, please contact the Director of the Office of Disability Services at **303-458-4941** for discussion of disability issues. Since accommodations are not retroactive, it is suggested that you contact the Disability Services Office as soon as possible; adequate lead-time may be required to receive accommodations, including review of appropriate documentation.

**Academic Dishonesty**
Regis University is committed to intellectual integrity in its academic pursuits. Sanctions may, therefore, be imposed by the College for Professional Studies faculty or schools for cheating (defined as using inappropriate sources of information on a test) or plagiarism (defined as presenting as one's own, the ideas, words, or products of another). Such sanctions may include a failing grade on the assignment, failure of the course, or expulsion of the student from the course or the department.

**Non-Public, Confidential or Proprietary Information Compliance**
For class projects that relate to a student's employment, the student is responsible for complying with all applicable policies of the employer, including those related to non-public, confidential or proprietary information